

Jay Joshi

B.Tech CSE Student

Specializing in Artificial Intelligence and Data Science, with experience in machine learning, computer vision, and data analysis. Hands-on with Python, OpenCV, YOLO, and Scikit-learn across academic and project work. Strong fundamentals in software development and analytics.

Projects

AI Based Project

Real-Time Motion Detection Using Open CV

- Built an end-to-end computer vision pipeline using YOLOv8 and Deep SORT, enabling real-time multi-object detection and persistent tracking across video frames.
- Developed a Flask-based visualization interface and optimized inference pipeline to achieve near real-time performance on standard hardware.
- Structured the project for modularity and reproducibility, with full source code and documentation available on GitHub.
- Achieved 20 FPS on standard hardware.

Active Development

System-Aware ML Optimizer

- Hardware-aware PyTorch model profiler that auto-detects CPU/GPU/Apple Silicon MPS, benchmarks latency and memory, and recommends optimal INT8/FP16 configurations via a Streamlit GUI and fault-tolerant CLI
- Bridges deep learning infrastructure and production optimization by dynamically assessing latency constraints and autotuning compressed models based on the host system's physical capabilities
- Uses **tracemalloc** to measure actual intermediate tensor footprints, dynamically cutting off benchmarks once inferences reach a coefficient of variance below 5%
- Automatically traverses nested layer blocks applying INT8 quantization across Conv1d, Conv2d, Conv3d, LSTM, GRU, and Linear perceptrons

Academic Project

Positive AI Companion & Chatbot

- Developed a responsive, browser-based chatbot using JavaScript and AI/NLP integration to provide emotional support and mood tracking for users.
- Designed an interactive "Mood Selection" onboarding flow that adapts the conversation logic based on real-time user sentiment (Happy, Stressed, Tired, etc.).
- Optimized frontend performance and accessibility, ensuring a seamless, low-latency conversational experience across mobile and desktop devices.

Noida, India

Mobile: [+91 73023 65820](tel:+917302365820)

E-mail: jay05.joshi@gmail.com

LinkedIn: [linkedin.com/in/gaminbhoot](https://www.linkedin.com/in/gaminbhoot)

GitHub: github.com/gaminbhoot

Portfolio: www.jayjoshi.online

SKILLS

Programming: Python, Java, JavaScript, SQL

Data Science & ML: Pandas, NumPy, Scikit-learn

Computer Vision: OpenCV, YOLOv8, Deep SORT

Web & Deployment: React, Vite, Flask, PHP

Tools: Git, Docker, VS Code

EDUCATION

Amity University

Noida, India

2023-2027

Bachelor's in technology
(Computer Science)

IIT Guwahati

Noida, India

2025-2026

Minor in data science

Vishwa Bharati Public School

Noida, India

2021-2023

Senior Secondary School, CBSE

CERTIFICATIONS

- NPTEL - Python for Data Science
- Cisco Networking Academy-CCNA: Introduction to Networks